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SUPERSEDING
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MILITARY SPECIFICATION

BOOTS, SKI-MOUNTAIN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a black ski-mountain boot.

1.2 Classification. The boots shall be of one type in the sizes and widths as specified in table I (see 6.2).

TABLE I. Schedule of sizes

Schedule of widths	5	5½	6	6½	7	7½	8	8½	9	9½	10	10½	11	11½	12	12½	13	13½
AA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
C	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
D	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
E	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
EE	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
EEE	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8430

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-203	- Paper, Kraft, Untreated
C-F-206	- Felt Sheet: Cloth, Felt, Wool, Pressed
L-P-390	- Plastic Molding and Extrusion Material, Polyethylene and Copolymers, Low, Medium, and High Density
V-L-61	- Laces, Nylon
V-T-285	- Thread, Polyester
V-T-295	- Thread, Nylon
KK-I-570	- Insole, Footwear, Leather, Cattlehide
KK-L-165	- Leather, Cattlehide, Vegetable Tanned and Chrome Retanned, Impregnated: and Soles
KK-W-231	- Welting, Leather Shoe
TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings
CCC-C-443	- Cloth, Duck, Cotton (Single and Plied Filling Yarns, Flat)
DDD-T-86	- Tape, Textile; Cotton, General Purpose (Unbleached, Bleached, or Dyed)
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-L-3122	- Leather, Cattlehide, for Footwear Uppers, Chrome Tanned, Fatliquored
MIL-C-13924	- Coating, Oxide, Black, for Ferrous Metals
MIL-L-27885	- Last, Footwear FWU-5/P
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-L-40051	- Leather, Cattlehide, for Glove Leather
MIL-L-41814	- Counter, Footwear

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STANDARDS

FEDERAL

FED-STD-151	- Metals; Test Methods
FED-STD-311	- Leather, Methods of Sampling and Testing
FED-STD-601	- Rubber: Sampling and Testing
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-731	- Quality of Wood Members for Containers and Pallets

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 36	- Brass Plate, Sheet, Strip, and Rolled Bar
D 5	- Test Method for Penetration of Bituminous Materials
D 1630	- Test Method for Rubber Property - Abrasion Resistance (NBS Abrader)
D 3951	- Standard Practice for Commercial Packaging
E 28	- Test Method for Softening Point by Ring-and-Ball Apparatus

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this document may appear in the sample, in which case this document shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.3 Materials.

3.3.1 Leather. All leather components covered by this document shall contain a paranitrophenol fungicide in conformance with the referenced leather documents.

* 3.3.1.1 Upper leather. The leather for the tip vamps, quarters, gussets, and backstays shall be smooth plated, full grain, or corrected grain side leather conforming to treatment A and treatment B of MIL-L-3122, except that the thickness shall be as specified for the cut parts in 3.7.1. The color shall be black. All finishes applied by the boot manufacturer and tanner shall be compatible with each other to ensure a long-lasting lustrous finish. The water resistance shall be obtained by the use of silicone treatments during tannage.

3.3.1.2 Top band.

3.3.1.2.1 Leather. The top band shall be cut from full grain leather conforming to type I, class a or b of MIL-L-40051 except the leather may be toggled or pasted, and the requirements for elongation shall not apply. The color shall be black. The leather shall be smooth finished on the grain side. The finish may contain dye and pigments only in amounts sufficient to provide a uniform color. The top band leather shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.4.1.1.

3.3.1.2.2 Cushion. The top band cushion shall be foam rubber, 3/8 inch in width and 5/16 inch in thickness.

* 3.3.1.2.3 Rubber pads (ankle binders). The ankle binders shall be foam rubber, 2 inches wide and 3/8 inch thick in the center tapering to a feather edge and shall extend from edge of backstay to the eye-row stitch.

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3.3.1.3 Linings. The lining leather shall be cut from full grain leather conforming to type II, class a or b of MIL-L-40051, except that the leather may be toggled or pasted and the requirement for elongation shall not apply. The color shall be tan. The leather shall be smooth finished on the grain side. The finish may contain dye pigments only in amounts sufficient to provide a uniform color. The lining leather shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.4.1.1.

3.3.1.4 Counters. The counters shall conform to the requirements of MIL-C-41814 for "Boot, ski, mountain." The counters shall be full molded conforming to the FWU-5P Last, and the bottom flange shall be molded to provide for full welt construction completely around the heel seat periphery of the boot.

* 3.3.1.5 Welting. The welting shall be 9/16 inch wide and conform to the requirements of KK-W-231.

* 3.3.1.6 Insoles, leather. The leather insoles shall conform to type I, tannage a or b, class 1 of KK-I-570. The insoles shall be 6 1/2 irons in thickness. The insoles shall have either a channeled (see 3.7.5.3) or stuck-on rib (see 3.3.6.6).

3.3.1.7 Midsoles, leather. The midsoles shall be full length, full grain leather conforming to types I, III or IV, class 3, style 1, sub-class b of KK-L-165. The midsole shall be 5 irons in thickness after being leveled.

3.3.2 Rubber compounds.

3.3.2.1 Midsoles, rubber. The rubber midsoles shall be made of natural rubber, synthetic rubber, or a mixture of natural and synthetic rubber. The rubber compound for the midsoles shall meet the following physical properties when tested as specified in 4.4.1.1. The midsole shall be 3 to 3 1/2 irons in thickness. The color shall be black.

a. The hardness shall be 90 ± 5. After accelerated aging for 70 hours at 212° ± 2°F, the hardness shall change no more than 5 points from the original.

b. The cut growth after accelerated aging for 70 hours at 212° ± 2°F shall be not greater than 500 percent after 25,000 cycles.

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* 3.3.2.2 Outsole and heel, rubber. The outsole and heel shall be a solid combined unit sole/heel and shall be made of natural rubber, synthetic rubber, or a mixture of natural and synthetic rubber. The rubber compound for the outsole and heel shall meet the following physical properties when tested as specified in 4.4.1.1.

a. The color shall be black nonmarking, producing only a slight erasable mark on white paper.

b. The abrasive index shall be not less than 80 before aging and shall not change more than 25 percent after aging 70 hours at 212°F ± 2°F.

c. The hardness shall be 80 ± 5. After accelerated aging for 70 hours at 212° ± 2°F, the hardness shall change no more than 10 points from the original.

d. The cut growth after accelerated aging for 70 hours at 212° ± 2°F, shall be not greater than 500 percent after 25,000 cycles.

* 3.3.2.3 Bottom design. The bottom shall be a Vibram lug design or equal. The outsole thickness including lugs shall be not less than 0.460 inch nor more than 0.545 inch, and excluding the lugs, the outsole thickness shall be not less than 0.145 inch nor more than 0.225 inch. The heels shall be a minimum 7/8 inch in height at the back center and a minimum 5/8 inch in height at the center breast.

3.3.3 Wedges, shank, polyethylene. The wedges shall be 6 to 7 irons in thickness conforming to type I, grade 2, class L, of L-P-390. The color shall be black.

3.3.4 Metal.

* 3.3.4.1 Nails, sole reinforcing. Sole reinforcing nails shall be brass conforming to Alloy No. 6 or No. 8 of ASTM B 36 and shall be commercial type No. 19 or "A" head and of sufficient length to produce a smooth "J" clinch on the insole.

3.3.4.2. Tacks and staples. Tacks or staples for attaching shank shall be of brass, shall be of sufficient length to thoroughly attach the parts through which they are driven, and shall leave the insoles smooth on the inside. Tacks shall be cut from material conforming to Alloy No. 6 or No. 8 of ASTM B 36.

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3.3.4.3 Eyelets and hooks.

3.3.4.3.1 Eyelets. The eyelets shall be commercial No. AA, invisible brass (plain), and of sufficient length to securely clinch.

* 3.3.4.3.2 Hooks. The hooks shall be solid brass and shall have a hollow post. The contractor shall submit the item proposed for use in the manufacturer of boots to the contracting officer for approval.

3.3.4.3.3 Finish of eyelets and hooks. After fabrication, the eyelets and hooks shall be coated with a standard nickel plating finish.

3.3.5 Shank combination. The shank combination shall be made of a shank board piece with single reverse rib steel shank securely attached by four prongs or two rivets. The size of the shank combination shall be in accordance with commercial practice.

3.3.5.1 Shank board. The shank board shall be a water-resisting type and shall have a finished thickness of 6-1/2/64 to 9-1/2/64 inches. The shank board piece shall be cut straight and shall be of a length and width to fill the heel seat cavity from approximately the breast of the heel to the heel seat lasting allowance. The shank board piece shall have a 5/8 ± 1/8 inch scarf at the breast of the heel, but no scarf at the extreme heel end.

3.3.5.2 Steel shank. The steel shank shall be constructed from 19 gage, 0.0418 ± 0.0003 inch cold rolled carbon steel with a hardness ranging from 47 to 54 Rockwell C scale when tested as specified in 4.4.1.1 and shaped to conform to the arch of the FWU-5P last. The width of the shank shall be 0.625 inch (+0.003, -0.007 inch). The shank shall be made with a single rib, and the overall thickness of the steel shank with rib shall be 0.080 to 0.125 inch. The steel shank shall have a zinc electroplated coating with dull or bright finish, or a zinc compound coating completely and uniformly applied to the base metal following a cleansing by any method of TT-C-490. The use of steel shanks with black oxide coating conforming to class 1 of MIL-C-13924 may be used in lieu of zinc compound coating. The rib shall taper off at a point within 1/2 ± 1/8 inch from the toe end and at a point within 1 ± 1/8 inch from the heel end.

3.3.5.3 Shank fitting sizes. The assembled pieces shall grade properly between sizes in accordance with commercial practice. In locating steel shank to shank board, the heel end of steel shank shall be located 1 1/8 inches ± 1/16 inch from heel end of shank board on all sizes. The shank casing schedule shall be in accordance with commercial practice.

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3.3.6 Fabrics.

3.3.6.1 Insole reinforcement. The insole reinforcement shall be a duck cotton cloth conforming to type I of CCC-C-443.

3.3.6.2 Backseam tape. The tape shall conform to type I, class 1, 1/2 inch width of DDD-T-86.

3.3.6.3 Felt insoles. The felt insoles shall conform to type I, class 12R3, 3/8 inch pressed wool felt of C-F-206.

3.3.6.4 Box toes. The box toes shall be cut from a material consisting of a woven base flannel or processed (needled) fabric impregnated with a pyroxylin or polystyrene compound. The box toes may be either the activated or prepared type. The thickness of this material shall be 0.0875 ± 0.0075 inch. The box toes shall be of a shape and size to engage the insole around its toe radius after lasting and shall be skived 9/16 ± 1/16 inch at the breast.

3.3.6.5 Laces. The laces shall conform to type II, class 3 of V-L-61. The color of the laces shall be black. The length of each lace shall be a minimum of 54 inches.

* 3.3.6.6 Stuck-on rib. The finished rib shall be 15/64 inches, ± 1/64 inch high and a minimum of 5/8 inch in width measured from the inside vertical portion of the rib, and shall extend around the periphery of the insole to provide for a welted heel seat. The rib shall be made from coated fabric or a combination of coated fabric and fiberboard. The fabric shall conform to requirements specified in 3.3.6.6.1. The upstanding portion of the rib shall be one of the following: not less than three plies of coated fabric folded over itself; two layers of coated fabric with a fiberboard filler not less than 0.050 inch thick; or fiberboard and coated fabric.

* 3.3.6.6.1 Rib construction. The ribbing fabric used in the construction of the rib shall be unbleached cotton or cotton-synthetic fabric conforming to the requirements of table II. The fabric shall be coated on one side with a rubber-base adhesive specified in 3.3.9 and bonded to the flesh side of the insole. The fabric or fiberboard shall cover 5/32 (-2/32, +1/32) inch of the peripheral edge and extend over the top of the vertical rib, or the fabric may extend from the inside top of the vertical rib and cover the area between the ribs for a distance not less than 5/8 inch from the inside rib edge. The preformed rib shall be firmly and smoothly fitted and bonded to the flesh side of the insole as specified above to provide the required extension on the finished boot.

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TABLE II. Ribbing fabric requirements

Characteristic	Requirement Minimum
Weight	8.0 ounces per sq. yd.
Yarns per unit length:	
Warp	52 yarns per inch
Filling	30 yarns per inch
Breaking strength:	
Warp	105 pounds
Filling	105 pounds

* 3.3.6.6.2 Rib strength. The physical requirements for the rib strength shall conform to table III when tested as specified in 4.4.2.4.

TABLE III. Rib strength requirements

Characteristic	Minimum (pounds) 1/	Average (pounds)
Shear strength	70	75
Stitch tear	20	30

1/ No single determination (sample unit) shall fall between the minimum value specified, and the average of all specimens (lot average) shall be not less than the average specified.

3.3.7 Thread.

3.3.7.1 Thread, upper fitting. All upper fitting operations shall be stitched with nylon thread conforming to type I or II, class 1 or type III, sizes FF and number 5 of V-T-295 except that requirements for colorfastness shall not apply. The color of the thread shall be black.

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3.3.7.2 Thread, sole stitching. The thread for sole stitching shall be polyester. The color of the running thread shall be black, and the color of the shuttle thread shall be natural. The polyester thread shall be size 10, 3 ply for the running thread and size 10, 3 ply or size 10, 6 ply for the shuttle thread and shall conform to type I, class 1, subclass C of V-T-285. The requirements for colorfastness shall not apply.

3.3.7.3 Thread, inseaming. The inseaming thread shall be polyester, size 10, 3 or 6 ply conforming to type I, class 1, subclass C of V-T-285. The color shall be natural.

* 3.3.7.4 Thread, stuck-on rib tape. The thread shall be natural color nylon conforming to type I, II, or III, class A, size E of V-T-295.

3.3.8 Bottom filler. The bottom filler shall be either thermoplastic or cold process type.

3.3.8.1 Thermoplastic bottom filler. The thermoplastic type shall consist of a mixture of ground cork and a suitable thermoplastic binder in proportion of 2 3/4 to 3 parts by volume of cork to each one part of binder. The cork granules shall be free from bark. The ground cork and the binder shall be thoroughly and evenly mixed. The binder shall be water insoluble. The binder shall have a softening point of at least 125°F and a maximum penetration of 85 mm with a 200 gram load for 60 seconds at 77°F when tested as specified in 4.4.1.1.

3.3.8.2 Cold process bottom filler. The cold process type shall be spreadable without the use of heat. It shall consist of a mixture of ground cork and a suitable binder in proportion of two parts cork to one part binder by volume. The cork granules shall be free from bark. When spread filler is dry and set, it shall consist of four parts cork to one part binder. The binder shall be water insoluble and have a softening point of at least 150°F when tested as specified in 4.4.1.1. The binder shall be tested after evaporation to a constant weight level.

3.3.9 Adhesive. The adhesive used for bonding the various parts of the boot shall be of the following types.

a. Natural rubber latex
b. Synthetic rubber latex (including chloroprene)
c. Natural rubber solvent cement
d. Synthetic resin cements
e. Synthetic rubber solvent cement

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3.3.10 Wax for inseaming and sole stitching. The wax shall be white or golden and shall be a permanently plasticized resin. The wax when in the temperature range of normal machine use shall thoroughly wax the thread for stitching through the midsoles, wedge, welt, and inseam.

3.3.11 Ink, edge setting. The ink shall produce a bright hard finish and shall be uniform in appearance after setting with a hot iron. The color of the ink shall be black.

3.3.12 Repairers. Repairers shall be liquid spray, crayon, or paste type applied by hand. The color shall match the upper leather and shall have sufficient coverage to correct minor surface imperfections of the leather.

3.3.13 Renovators. Renovators used in lieu of or in addition to repairers shall match the color of the upper leather. Application may be by sponge or spray method.

3.3.14. Fillers. Fillers, where used, should be capable of providing a foundation for application of the top finish and may be applied by sponge or spray method.

3.3.15 Top finish. The top finish may be applied by sponge or spray method and shall be a material compatible with and capable of wetting into and binding the previously applied coats. The finish shall match the dyed leather and shall provide a luster.

3.4 Design. The design shall be for a laced-to-toe blucher style boot with black grain-out upper leather, rolled cushion top, ankle binder pads, full gusset, one piece vamp, and lower quarter with the vamp extending to the gusset approximately 3 inches from the blucher points, and quarters closed at the back and reinforced with a full backstay that is looped at the top. The boot shall be fully lined with tan color glove type leather except that the gusset has no lining. The construction shall be all around Goodyear welt, with leather and rubber midsoles and polyethylene shank wedge. The outsole and heel shall be black rubber with a cleated Vibram design or equal. The boot has a removable felt slip sole for bottom insulation.

* 3.5 Patterns. The upper patterns shall be the contractor's commercial type and shall be approved by the contracting activity (see 6.2). The height of the finished boot, measured upward on outside from tread area at breast of heel to top of boot, shall be 9 inches ± 1/4 inch on size 9 and shall grade up or down between sizes according to contractor's commercial patterns. The contractor shall furnish the wood insole rounding patterns conforming to paper patterns loaned by the contracting activity. As an alternative, the insole patterns may be altered for fit to the periphery of the last bottom edge when the insole rounding method is not used.

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3.6 Lasts. The boots shall be lasted on the FWU-5P last of MIL-L-27885. All necessary sizes will be loaned to the contractor by the Government.

3.7 Construction.

3.7.1 Cutting uppers. The uppers shall be cut from grain-out leather. The leather shall conform to the thickness specified in table IV. No parts cut off stretch shall be accepted.

TABLE IV. Thickness of upper leather

Part	Thickness (ounces) Minimum	Maximum
Tip vamp	5	6
Quarter	4 1/2	6
Gusset	3	4
Backstay	4	6

3.7.1.1 Skiving. Each scarf shall be sufficient to provide a smooth even fitting. All skiving shall be on the flesh side.

3.7.1.2 Uppers. The quarters shall be skived at the bottom edge and at the blucher nose and tip vamp edge. The tip vamp shall be skived at the bottom edge, vamp and throat. The gusset bottom edge shall be skived. Both ends of the backstay loop shall be skived.

3.7.1.3 Linings. The quarter lining shall be skived at the bottom, top, back, and front egdes. The vamp lining shall be skived at the bottom, top, back, and tip vamp edges.

3.7.2 Contractor's marking. The contractor's name, number and date of contract, purchasing office, and correct size and width of boot shall be legibly stamped with marking ink in the center of the outside quarters on the quarter lining just below the top band on each boot.

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3.7.3 Marking, permanent identification. The inside quarter of each boot shall be marked on the grain side with the correct size and width, the contractor's identification symbol, and the month and year (expressed numerically) of the date of contract. This marking shall be impressed into the grain side of the leather in such a manner as to be permanently visible without cutting through at any point, shall be placed approximately in the center of the inside quarter, and shall be placed just below the top band. Figures shall be Arabic and letters Gothic. These figures and letters shall be a minimum of 9/32 inch and maximum of 3/8 inch in height. The contractor's symbol shall be in block as shown by the following example.

10 E AB 9-86

* 3.7.4 Upper fitting. Foam ankle binders shall be cemented to the flesh side of the linings. The ankle binders shall be positioned 2 1/4 inches (± 1/8 inch) from the top of each quarter. The linings shall be cemented to the quarters. The upper fitting stitch types, thread sizes, and stitches per inch shall conform to the requirements specified in table V. The stitch and seam types specified in table V shall conform to FED-STD-751.

TABLE V. Stitch and seam types

Location	Stitch Type	Number of stitches per inch	Thread size
Quarter lining closing (toe area)	304	14-16	FF
Quarter and quarter lining backseam closing	304	14-16	FF
Backseam tape	301	8-10	FF
Top band	301	8-10	FF
Vamp closing (back)	304	14-16	FF
Loop and attach backstay	301	8-10	FF
Gusset and eyelet row	301	8-10	FF
Vamp to gusset and quarters attaching (2 rows)	301	8-10	FF
Vamp to gusset and quarters (reinforcement - 1 row)	301	4-5	5

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3.7.4.1 Quarter lining closing toe area. Both edges of lining shall be butted at toe and seam as specified in table V. Width of stitching shall be 3/16 to 1/4 inch. Seam shall be rubbed flat.

3.7.4.2 Quarter and quarter lining backseam closing. The quarters shall be superposed with the edges of each quarter and quarter lining butted and seamed. Width of stitching shall be 3/16 to 1/4 inch. Seam shall be rubbed flat.

3.7.4.3 Backseam tape. The backseam tape shall be centered over the quarter closing and seamed with two rows of stitches 3/8 inch apart centered on the tape.

3.7.4.4 Top band. The top band shall be superposed grain to grain on top edge of quarters and seamed with one row of stitches 1/8 inch from the edges catching the edge of backseam tape. A foam collar shall be cemented to the top of the quarters and the leather top band rolled over the top of the foam collar and cemented, prior to stitching. The top band shall be turned over edges of quarters and quarter lining and shall be seamed with one row of stitches 1/8 ± 1/32 inch from the edge.

3.7.4.5 Vamp closing (back). The back edges of the one piece vamp shall be butted and seamed with one row of stitches 3/16 to 1/4 inch wide. Seam shall be rubbed flat.

* 3.7.4.6 Loop and attach backstay. The backstay shall be inverted, and 1 5/8 to 1 3/4 inches of the narrow end shall be superposed squarely over upper end of taped backseam closure with grain to grain. This shall be stitched to boot with 5/8 by 1 1/2 inch box stitch with top of box stitching located 1/2 ± 1/32 inch below top of quarters. The backstay shall be turned down over the backseam closure, leaving a loop extending up to 1/2 inch above the top of the quarters. Stitching shall be up one side of backstay to within 1 1/2 inches of top of quarters, across backstay, and down the other side with two equally spaced rows of stitches 1/16 inch apart with the outside row of stitching 1/32 to 1/16 inch from the edges of the backstay. One row of heavy duty lock stitches shall run on each side of backstay from welt upward 3 inches ± 1/8 inch.

3.7.4.7 Gusset and eyelet row. The gusset shall be placed along one edge of the quarter beginning at the top and seam 1/16 inch from the edge to bottom and continue with another row of stitches spaced 3/4 inch from the first row and then finish across top to the first row of stitches. The other edge of gusset shall be joined to quarter in the same manner.

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3.7.4.8 Vamp to gusset and quarters attaching. The tongue portion of the vamp shall be positioned to proper location on the gusset and cement. The vamp shall be positioned to quarters and seamed with two rows of stitches 1/8 inch apart, with outer row 3/32 to 1/8 inch from raw edge of vamp.

3.7.4.9 Vamp to gusset and quarter reinforcement. One row of stitches shall be placed 5/32 to 3/16 inch below the two 1/8-inch apart rows of stitches when joining vamp to gusset and quarters.

* 3.7.4.10 Hooks and eyelets. All sizes of boots shall have nine eyelets (as specified in 3.3.4.3.1) in each quarter with additional stud hooks (as specified in 3.3.4.3.2) in four eyelets beginning with the second eyelet from the top of each quarter. The edge of eyelets or hooks shall be 3/16 inch (+ 1/32, - 1/16 inch) from the edge of the quarter. The eyelets and hooks shall be spaced evenly from blucher nose to top of quarters. The eyelets shall be smoothly and securely clinched.

3.7.4.11 Lacing for lasting. The machine lacing for lasting shall provide for a 1 ± 1/8 inch opening after lasting. The quarter shall be laced in the lower three eyelets.

3.7.5 Sole leather and stock fitting. The forepart of the wedges shall be skived 5/8 ± 1/8 inch and shall extend from the ball line to the end of the heel seat. Prior to the assembly of the wedge, midsoles leather, and rubber, the grain side of the leather midsole and the reverse side of the rubber midsole and mating side of the wedge shall be roughed their full length with a fine wire brush or coarse abrasive paper. Two coats of chloroprene adhesive as specified in 3.3.9 shall be applied to the roughed grain side of the leather midsole. The first coat serves as a primer and shall be allowed to dry before application of the second coat. A single coat of the same adhesive shall be applied to the mating surface of the rubber midsole and wedge. After assembled wedge and midsoles have been properly tempered and molded, they shall be laid on a sole laying machine with pressure.

3.7.5.1 Insoles. Insoles shall be sorted for varying fibers to get uniform channeling and casing for even weight. The insoles shall be fleshed. The insoles shall be rounded to pattern conforming to FWU-5P last bottoms. Die cut insoles will be accepted provided they conform to FWU-5P last bottoms.

3.7.5.2 Insole marking. Each insole shall be stamped in the center of the shank on the grain side with the applicable size and width, using figures for the whole sizes and letters for the widths. The stamping must be plainly visible with the letters and figures at least 1/4 inch in height and the stamped impression legible.

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* 3.7.5.3 Insole channeling. The insoles, except stuck-on rib insoles, shall be channeled all around and may be chamfered. The channel leaf and outside lip shall be not less than one-third the thickness of the insole, and the channel leaf shall be slightly thicker than the lip. The in-between substance shall be 1/64 to 1/32 inch greater than the combined thickness of the leaf and lip. The channel margin shall be a minimum of 5/32 inch, and the maximum shall not exceed the dimensions necessary to achieve the required extension of the finished boot as specified in 3.7.15. The lips shall be cemented with a latex adhesive specified in 3.3.9 (a) or (b) and pressed together at lip setting to incline inward and be a minimum of 1/8 inch in height.

* 3.7.5.3.1 Assembly of stuck-on rib insoles. When assembled, the stuck-on rib insoles shall meet the requirements of 3.3.6.6. In addition, after applying the stuck-on rib to the flesh side of the insole, the 3/4-inch tape made from the duck cloth specified in 3.3.6.1 shall be stitched, with the nylon thread specified in 3.3.7.4, 3/16 ± 1/16 inch from the tape edge with two rows of stitching, 3/16 ± 1/16 inch apart, using 5 to 7 stitches per inch.

* 3.7.5.4 Insole backing. The insoles, except stuck-on rib insoles, shall be reinforced by the application of cotton duck as specified in 3.3.6.1. The duck shall be coated on one side with any adhesive compound as specified in 3.3.9 and shall be bonded to the flesh side of the insole. The duck shall cover the entire area between the ribs from the heel end forward and extend to the top of the upstanding rib, shall be firmly and smoothly fitted and bonded to the base and the lip, and shall be trimmed to the rib height. The insoles shall be cemented with any adhesive as specified in 3.3.9 on the right flesh side prior to attaching the coated duck. The 3/4-inch tape reinforcement method may be used as an alternative, utilizing the duck conforming to 3.3.6.1.

3.7.6 Lasting. Uppers may be conditioned by any suitable means except they shall not be dipped into water. The correct size and width of uppers, lasts, insoles and spacers shall be assembled in a manner to provide for a welt around heel seat. Prior to tacking the insoles to the last, a spacer shall be used to allow for the felt insole. The correct size felt insole as specified in 3.3.6.3 may be utilized as the spacer, or its equivalent, and shall be placed between the insole and last. Insoles shall be tacked to the spacers through to the last with not less than five tacks: one of the five tacks shall be in the center of the heel seat, one at shank, one at each side of the ball, and one at the toe. Tacks shall be of sufficient length to thoroughly attach the parts through which they are driven. Edges of insole shall not extend beyond the last bottom at any point. Counters specified in 3.3.1.4 shall be assembled in accordance with sizing schedule so as to effect full inseam stitching. When leather counters are used, the flesh side of the

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leather counter shall be towards the inside of the boot and inserted between quarters and lining. The correct size of pyroxylin or polystyrene type box toe specified in 3.3.6.4 shall be conditioned and inserted between the tip vamp and lining. Uppers shall be drawn over the lasts with tension on the pulling-over machine to ensure that the quarters at the blucher points and the vamps are down to the last, blucher noses are even, and box toes are aligned and mated for length. Sides of boots shall be spindled, drawing the upper snugly to the last, and then side-lasted so that, when stapled, the upper will be held securely to the last. The lining shall be pulled smooth and tight. The heel seat shall be wiped-in and wire or nylon filament attached securely around the heel insole rib and shall be free from wrinkles. The toe shall be smoothly wiped-in and the toe wire or nylon filament attached securely around the toe at the base of the insole rib.

3.7.6.1 Time allowance on lasts. The boots shall remain on the lasts a sufficient length of time to thoroughly dry the counters, uppers, and sole leather.

3.7.7 Inseaming. Inseaming shall be done by machine, with hot waxed thread and with tension on the thread, using a needle not larger than number 41 and with not less than 3 1/4 stitches to the inch. The welting shall be tempered to provide a tight seam. The welting shall be inseam stitched to the bottom of insole rib completely around the periphery of the boot to provide for a welted heel seat.

3.7.8 Tack pulling, inseam trimming. All tacks shall be removed, and no broken tack points shall remain. The inseam shall be closely trimmed all around without cutting or damaging the stitches. The welt shall be beaten out smoothly while in temper and slashed or stretched around the toes and heel seats.

3.7.9 Shank fitting - bottom filling. The shanks shall be selected for size as specified in 3.3.5.3. The shanks shall be inserted in position with the shank board piece extending to the heel seat cavity and the rib of the shank centered to the contour of insole bottom. The shanks shall be attached with two brass tacks or staples as specified in 3.3.4.2, one at each side of the shank board back of the breastline. The forward end of the shank shall lie flush with the insole and fit the contour of the boot bottom rearward of the ball line. The bottom filler shall be firmly pressed into the insole channel around the toe and shall extend with a uniform even surface to the forward end of the shank heel seat filler. The bottom shall present a flat smooth surface for sole laying. No bottom filler shall remain under the shank piece or over the welt. Any excess cavities at the heel portion shall be filled with bottom filler.

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3.7.10 Sole laying, midsoles and wedges. The boot bottoms shall be thoroughly cemented with any adhesive specified in 3.3.9, and after the assembled wedge and leather rubber midsoles have been cemented and molded as specified in 3.7.5, the leather midsole shall be cemented over the entire inner surface and shall be laid on a sole-laying machine with pressure. The soles shall be of adequate size and shall be laid evenly to get the specified edge extension.

3.7.11 Rough rounding. The forepart and heel seat of the combined midsoles shall be smoothly rounded on a rough rounding machine for the specified edge extension (see 3.7.15). The sides of the forepart from ball line to corner of toe shall be straight with adequate extension for laying the cleated rubber outsole.

3.7.12 Sole stitching, midsoles and wedges. Prior to attaching the cleated rubber outsole, the wedge and leather and rubber midsoles shall be stitched to the welt on a lockstitch machine using polyester thread, a No. 45 round blade chisel point awl, not larger than a No. 45 needle, and 5 1/2 to 7 1/2 stitches per inch. The stitching shall be laid on the surface of the welt and close to the outer edge of the welt on the finished boot. The stitching shall extend around the heel seat, close to the upper, in order to eliminate any chance of cutting the stitch in the heel grooving operation. The stitching shall be with the lock (peg) just under the surface of the rubber midsole.

* 3.7.13 Leveling. Prior to attaching the cleated rubber outsole, the bottoms shall be progressively level while in temper on a leveling machine by means of metal rolls. The machine shall be adjusted to level the bottom of the midsole from the toe back to heel with sufficient pressure to the shank edges for a tight seam. As an alternative, any suitable equipment may be used that will adequately level the sole bottom while in temper.

3.7.14 Outsole attaching. The rubber midsole and the reverse side of the cleated outsole shall be buffed their full length with a fine wire brush or coarse abrasive paper, then thoroughly covered with a synthetic adhesive specified in 3.3.9. After sufficient drying, the cleated outsole specified in 3.3.2.2 and 3.3.2.3 shall be attached on a cement sole attaching machine with pressure to ensure a permanent adhesion. The outsole shall be of adequate size and fitted from the toe with the edge of the midsoles slightly overlapping the outsole.

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3.7.15 Edge trimming. Edges shall be trimmed flat all around, using a flat bed cutter or equivalent type cutter, to produce an edge allowing for the required finished extension of outside ball to be not less than 3/16 inch, the toe trimmed square to be not less than 1/4 inch, and the inside ball and around heel seat to be not less than 1/8 inch, when measured at right angles from the upper. The squared toe extension shall allow for adequate fitting of ski binding.

3.7.15.1 Edge setting. The leather portion of the edge, consisting of welt and midsole, shall be stained to match the color of the upper leather and set on the edge setting machine and made smooth and bright. The top surface of the welts shall be black. No iron stain shall be permitted.

3.7.16 Heel trimming, grooving, finishing. The heels shall be trimmed with a single horizontal groove, 3/16 ± 1/32 inch in depth and 5/8 ± 1/16 inch in width except at the heel breast where it shall be 3/4 ± 1/8 inch in width for a distance of 3/4 ± 1/16 inch. The grooving operation shall not cause damage to the heel seat stitching. The leather heel seat shall be stained and finished to match the sole edge.

3.7.17 Sole reinforcing. The soles shall be reinforced with brass No. 19, or A head nails specified in 3.3.4.1. The nailing pattern for size 9 shall be 13 nails placed in between the cleats inside the seam, with three nails in the toe area and five nails on each side of the inside and outside ball extending from the toe area to just below the ball line. The number of nails used on other sizes shall be in accordance with standard practice. The nailing machine shall ensure a 1/16 inch minimum, firm, smooth turned (J) clinch. The machine driver shall be set so that the nail heads will sink slightly below the surface of the outsole. No nails shall be driven through the inseam.

3.7.18 Finishing.

3.7.18.1 Preparation. The boot shall be cleaned by removing accumulated dirt such as wax or cement, and the surface of the leather shall be conditioned to receive further application of the finish. All thread ends shall be trimmed.

3.7.18.2 Treeing. All wrinkles shall be removed. No material shall be used that may injure the leather or thread.

3.7.18.3 Nails and tacks. Tacks and nails that have been left protruding through or around the insole and cannot be pulled out shall be cut close to the surface and shall leave no protruding stumps.

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3.7.18.4 Final finish. Boots shall be repaired and filled. All raw edges, including backstay and front edges, shall be stained to match the upper leather. Materials and methods shall be as specified in 3.3.12 to 3.3.14 inclusive.

3.7.19 Laces. The boots shall be mated. A lace shall be inserted through the lower eyelet of the outside quarter of each boot, and the two laces shall be tied firmly together.

3.7.20 Felt insoles. Two pairs of felt insoles of the correct size as described in 3.3.6.3 shall be supplied by the contractor and packed with each pair of boots. Felt insoles for lasting the upper, when used as a spacer, may be supplied to the Government as one of the two pairs, providing no damage has resulted to the insoles.

3.8 Workmanship. The finished boots shall conform to the quality and grade of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.2), it shall be examined for the defects specified in 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.1.1 Component testing. Testing shall be performed on components and materials listed in table VI. A rubber compound lot shall consist of the amount of rubber compound that is produced or compounded in one day. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in table VI, they shall be as required by the referenced test method. The lining leather sampling procedure shall be in accordance with FED-STD-311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units, the lot average, or the composite sample fail to meet any requirement specified. The sample size shall be as follows:

Lot size	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE VI. Component tests

Component and lot size expressed in terms of	Characteristic	Requirement para.	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. of det. per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Top band (leather) (1 band)	Paranitrophenol	3.3.1.2.1	6711 1/	-	X	-	-	-	5 bands
Cushion, top band	Material identification	3.3.1.2.2	2/	-	-	-	-	-	-
Rubber pad (ankle binder)	Material identification	3.3.1.2.3	2/	-	-	-	-	-	-
Lining leather	Paranitrophenol	3.3.1.3	6711 1/	-	X	-	-	-	4.4.1.1
Midsoles rubber compound (lot) (1 pair)	Hardness (before and after aging)	3.3.2.1	3021 and 7221 3/ 4/	-	X	5	-	Whole number	One midsole
	Cut growth (after aging)	3.3.2.1	2/	-	-	-	-	-	-
Outsole and heel compound (lot) (1 pair)	Abrasive index (before and after aging)	3.3.2.2	D 1630 5/ and 7221 3/ 4/	-	X	3	-	Whole number	One outsole and heel
	Hardness (before and after aging)	3.3.2.2	3021 and 7221 3/ 4/	-	X	5	-	Whole number	-

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TABLE VI. Component tests - Continued

Component and lot size expressed in terms of	Characteristic	Requirement para.	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. of det. per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Outsole and heel compound (lot) (1 pair) - continued	Cut growth (after aging)	3.3.2.2	2/	-	-	-	-	-	-
	Non-marking (sole and heel)	3.3.2.2	4.5.1	X	-	1	X	-	1 sole & heel
Nails, tacks and staples	Material identification	3.3.4.1 and 3.3.4.2	2/	-	-	-	-	-	-
Eyelets and hooks	Material identification	3.3.4.3.1 and 3.3.4.3.2	2/	-	-	-	-	-	-
	Finish	3.3.4.3.3	2/	-	-	-	-	-	-
Shank board	Material identification	3.3.5.1	2/	-	-	-	-	-	-
Shank steel (1 pair)	Material identification	3.3.5.2	2/	-	-	-	-	-	-
	Hardness (Rockwell C scale)	3.3.5.2	Method 243 6/	X	-	1	-	Hardness number	1 shank
	Coating identification	3.3.5.2	2/	-	-	-	-	-	-
	Shank width and thickness	3.3.5.2	2/	-	-	-	-	-	-

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TABLE VI. Component tests - Continued

Component and lot size expressed in terms of	Characteristic	Requirement para.	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. of det. per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Results reported as: Sample unit
Box toes	Material identification	3.3.6.4	2/	-	-	-	-	-	-
	Thickness	3.3.6.4	2/	-	-	-	-	-	-
Bottom filler (1 lb)	Material identification	3.3.8	2/	-	-	-	-	-	-
	Solubility of binder in water	3.3.8.1 and 3.3.8.2	2/	-	-	-	-	-	-
	Softening point of binder	3.3.8.1 and 3.3.8.2	E 28 5/	-	7/	2	-	Avg of 2 determinations to nearest 1°F	1 lb composite
	Penetration of binder	3.3.8.1	D 5 5/	-	7/	3	-	Avg of 3 determinations to nearest mm	1 lb composite
	Composition	3.3.8.1 and 3.3.8.2	2/	-	-	-	-	-	-
Adhesive	Material identification	3.3.9	2/	-	-	-	-	-	-
Wax	Material identification	3.3.10	2/	-	-	-	-	-	-
Ink	Material identification	3.3.11	2/	-	-	-	-	-	-

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TABLE VI. Component tests - Continued

1/	Refers to FED-STD-311.
2/	A certificate of compliance shall be submitted and will be acceptable for the stated requirement.
3/	Refers to FED-STD-601.
4/	Except time shall be 70 hours and temperature 212°± 2°F.
5/	Refers to ASTM standard.
6/	Refers to FED-STD-151.
7/	Requirements applicable to "lot average".

4.4.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

* 4.4.2.1 In-process examination of uppers after all fitting. The upper assembly shall be examined for defects in cutting, fitting, and other construction characteristics that cannot be seen in the end item. The defects shall be classified as in table VII. The lot size shall be expressed in units of uppers. The sample unit shall be one completely fabricated upper assembly prepared for lasting. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

NOTE: Defects designated by an asterisk shall be scored as major when seriously affecting serviceability and minor when affecting serviceability but not seriously.

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TABLE VII. Upper assembly defects

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship	Any component missing or not specified type	X	
	Any component misplaced or not affixed as specified	*	
	Backseam not rubbed down		X
	Quarters, blucher nose, tip vamp, gusset, and backstay not skived as specified		X
	Lining not skived as specified		X
Quality of leather	Thickness more than 1/2 ounce less than the minimum specified	X	
	Thickness less than specified minimum but not exceeding 1/2 ounce less		X
	Thickness more than maximum specified		X
	Off-stretch cuts	X	
	Slaughter cut	*	

* 4.4.2.2 In-process examination of boot before bottom filling. The partially fabricated boot shall be examined for defects in construction and workmanship characteristics that cannot be seen in the end item. Defects shall be classified as in table VIII. The lot size shall be expressed in units of boots. The sample unit shall be one partially constructed boot at a point after lasting and attachment of shank, but before application of bottom filler. The inspection level shall be II, and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE VIII. Boot (before bottom filling) defects

Examine	Defect	Classification	
		Major	Minor
Bottom of boot	Any component missing or not specified type	X	
	Shank not properly positioned or not fitting contour of boot bottom	*	
	Shank wrong size, malformed, or not securely attached		X
	Any tear in cotton duck in stitch area:		
	-more than 3/4 inch	X	
	-more than 1/2 inch but not more than 3/4 inch		X
	Any insole or anchor tack not removed	X	
	Upper damage	*	
	Poor heel seat, side or toe lasting	*	
	Inseam not properly trimmed	*	
	Less than 3 stitches per inch on inseam	X	
	Less than 3 1/4 stitches but not less than 3 stitches per inch on inseam		X
	Inseam stitches skipped, cut, or damaged	X	
	Inseam stitches not at bottom of insole rib or not in welt groove	*	
	Broken insole rib	X	
	Welts not beaten out and stretched or slashed around toe and heel		X
	Counter not caught in inseam stitching		X
Upper part of boot	Uppers not tightly pulled down to last		X
	Lace opening less than 7/8 inch or more than 1 1/8 inches		X
	Quarter not laced in three lower eyelets		X

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* 4.4.2.3 In-process examination of sole stitching before attaching rubber outsole. The partially fabricated boot shall be examined for defects in sole stitching characteristics that cannot be seen in the end item. Defects shall be classified as in table IX. The lot size shall be expressed in units of boots. The sample unit shall be one partially constructed boot at a point after stitching of the wedge, and leather and rubber midsoles, but before application of the rubber outsole. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE IX. Sole stitching defects

Examine	Defect	Classification	
		Major	Minor
Wedge and midsoles, leather and rubber, stitching	Less than 5 stitches per inch or more than 9 stitches per inch	X	
	Less than 5 1/2 but not less than 5 stitches per inch, or more than 7 1/2 but not exceeding 9 stitches per inch		X
	Skipped or broken stitch:		
	-seriously affecting serviceability	X	
	-affecting serviceability but not seriously		X
	Stitch locks laying on surface:		
	-seriously affecting serviceability	X	
	-affecting serviceability but not seriously		X
	Midsoles, wedge, or welt damaged by stitches	X	

* 4.4.2.4 In-process testing (stuck-on rib insole). When used, the stuck-on rib shall be tested for compliance with the requirements of 3.3.6.6.2. Method 2061 of FED-STD-311 shall be used for shear strength test, and Method 2171 of FED-STD-311 shall be used for stitch tear strength. The sample unit shall be two ribbed insoles, one for each test (without the support stitching), and the

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sample size shall be five, regardless of the lot quantity. Requirements are applicable to the sample unit for minimum unit requirements and lot average for average requirements. The lot shall be unacceptable if one or more sample units or the lot average of all units fail to meet the requirements specified. All test reports shall contain the individual values utilized in expressing the final results.

* 4.4.3 End item visual examination. The end item shall be examined for the defects listed in table X. The lot size shall be expressed in units of boots. The sample unit shall be one completely fabricated boot and the selection shall be by pairs. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor) combined defects, except for protruding point of nail or tack forward of heel breast line on insole. Defects of pairing shall be classified as a single defect.

NOTES: Appearance defects shall be scored only when the condition is plainly visible at a distance of 3 feet or more.

TABLE X. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Pairing	Not properly mated (i.e., not right and left of same size, variation in color or appearance)	*	
	Box toe crooked, long, or short	*	
	Variation of more than 1/4 inch in height of pair	X	
Cleanness	Any nonremovable spot, stain, or foreign matter affecting appearance		X
Color and finish	Not specified color	X	
	Finish streaky, chipped, or flaky affecting appearance		X
	Any raw edges not stained to match upper leather		X

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TABLE X. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Color and finish - continued	Sole edges or heels not bright or not finished as specified		X
	Heel seats or top of welts not finished as specified		X
Design and size	Not as specified	X	
Upper leather	Leather deeply snuffed (i.e., fiber structure damaged)	X	
	Any grub or tick damage, scratch, brand, or pipey or boney leather		*
	Fat wrinkles, flanky leather, or veins		*
	Stretchy vamp	X	
	Leather embossed or printed	X	
Construction and workmanship	Any cut, tear, hole, rip, repair, or factory damage		*
	Any component missing or assembly misplaced, operation omitted or not properly performed		*
	Wrinkled or bunched area at the back seam		*
Seams and stitching (upper)	Open seam		*
NOTE:	A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuous skipped or run-off stitches occur. On multiple stitched seams, a seam is considered open when either one or both sides of the seam are open.		

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TABLE X. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Seams and stitching (upper) - continued	Loose tension resulting in a loosely secured seam	*	
	Tight tension resulting in puckering or cutting of leather	*	
	Wrong stitch or seam type	X	
	Two or more stitches less than the specified minimum	X	
	One stitch less than the specified minimum		X
	More than the specified maximum number of stitches	*	
	NOTE: A plus tolerance of 3 stitches per inch will be allowed when stitching over heavy places or turning sharp corners.		
	Gage of stitching not as specified or irregular	*	
	Stitching omitted where required	*	
	Needle holes or needle chew	*	
	Thread ends not trimmed throughout boot		X
Counters	Rolled or curled counter	*	
	Soft counter		X
Edge making	Edge trimmed into sole stitching	*	
	Sole extension less than the specified minimum	*	
	Edge setting not as specified	*	
	Checked sole (i.e., separation between outsole and midsole)	*	
	Edge not properly trimmed		X

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TABLE X. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Eyelets and hooks	Number of eyelets or hooks not as specified but each row within the pair has the same number		X
	Not the same number of eyelets or hooks in each row	X	
	Edge of any eyelet or hook less than 1/8 inch or more than 7/32 inch from edge of quarter		X
	Any eyelet or hook not smoothly or securely clinched		X
	Eyelets not properly spaced within the row or misalignment between the rows to an extent interfering with proper lacing	X	
Inseaming	Grinning seam (i.e., thread exposed)	X	
	Strained seam (i.e., needle holes visible but thread not exposed)		X
Leather lining	Leather lining having excessive fullness, torn, loose, or wrinkled	*	
Insole	Short or long	*	
	Any protruding point of tack or nail in heel seat area	X	
	NOTE: Any protruding point of tack or nail on insole forward of heel breast line, found in sample, shall cause rejection of lot presented.		
Heel finishing and attaching	Heel not trimmed and grooved to the specified measurements		X
	Heel not finished smooth		X
	Checked heel (i.e., separation at heel and sole)		X
	Crooked or wrong heel size	X	

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TABLE X. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Heel finishing and attaching - continued	Any heel nail missing	X	
	More than one of three back nails not clinched on the insole	X	
	One of three back nails not clinched on the insole		X
	Any corner breast nail not clinched on the insole	X	
	More than one nail, other than corner breast or three back nails, not clinched on the insole		X
	NOTE: Evidence of brass on the insole shall indicate clinching.		
Felt insole (2 pairs)	Missing or wrong size	X	
Marking	Missing, incomplete, incorrect, not applied in the specified manner, misplaced, illegible or not specified size		X

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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Examine	Defect
Materials	Any component missing, damaged, or not as specified Any component damaged, affecting serviceability
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number of pairs of boots per container is more or less than required

* 4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Test for nonmarking soles and heels. The edge of the outsole or heel shall be drawn across a sheet of white bond paper using moderate pressure. If only a light mark is made which is easily erased by rubbing with fingers, the outsole or heel shall be classed as nonmarking.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each pair of properly mated boots shall be tied together by the ends of laces, which shall be inserted through the lower eyelet of the outside quarter of each boot. The felt insoles shall be placed inside each pair of boots (see 3.7.20).

* 5.1.2 Commercial. Boots shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

* 5.2.1 Level A. Twelve pairs of boots of one size and width only, unit packed as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style FOL-L, grade V2s of PPP-B-636. The inside of each fiberboard shipping container shall be fitted with a taped box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The boots shall be arranged on their sides in three layers of four pairs per layer, with each pair of boots placed toe to toe with the quarters overlapping and every other pair reversed (heel to toe). Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. The inside dimensions of the shipping container shall be as specified in table XI. Each container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4. Toward the end of the contract, or when there are less than the required amount per container of the same size and width, mixed sizes and widths may be packed in the same shipping container. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.2.2 Level B. Twelve pairs of boots of one size and width only, unit packed as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style FOL-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each fiberboard shipping container shall be fitted with a taped box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. The boots shall be arranged on their sides in three layers of four pairs per layer, with each pair of boots placed toe to toe with the quarters overlapping and every other pair reversed (heel to toe). Each container shall have the contents completely covered on the

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top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. The inside dimensions of the shipping container shall be as specified in table XI. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4. Toward the end of the contract or when there are less than the required amount per container of the same size and width, mixed sizes and widths may be packed in the same shipping container.

* 5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4.

TABLE XI. Approximate inside dimensions of shipping containers 1/

Range of boot sizes	Length (inches)	Width (inches)	Depth (inches)
5 AA through 9 1/2 EEE	36 1/2	13	13
10 AA through 13 1/2 EEE	37 3/4	14 1/2	15 1/4

1/ Approximate dimensions are furnished as a guide only. The dimensions of the shipping container shall be such that it will result in a snug fit on the contents to be packed therein. The use of blocking materials to accommodate small size boots in an oversized shipping container will not be permitted.

* 5.2.3 Commercial. Boots, unit packed as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), boots packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall

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be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern 2 shall be used for boot sizes 5AA through 9 1/2 EEE, and pallet pattern 94 shall be used for boot sizes 10AA through 13 1/2 EEE. Interlocking of loads shall be effected by reversing the pattern of each course.

* 5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Labels, mixed sizes and widths. Each shipping container packed with mixed sizes and widths shall have securely attached to the end and side, directly under the printing or stencilling, a white paper label 5 by 4 inches with the words "MIXED NSN's" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and NSN's contained therein.

6. NOTES

6.1 Intended use. These boots are intended for wear by personnel of the Department of Defense for mountain climbing and skiing.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Sizes and widths required (see 1.2).
c. When a first article is required (see 3.2, 4.3, and 6.4).
d. Approval of upper patterns by the contracting activity (see 3.5).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 Sample. For access to samples, address the contracting activity issuing the invitation for bids.

* 6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

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6.5 Subject term (key word) listing.

Boots, ski-mountain
Mountain climbing
Skiing

6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

Project No. 8430-0360

Review activities:

Army - MD
Air Force - 82, 11
DLA - CT

User activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-43739A	BOOTS, SKI-MOUNTAIN

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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